Physics 180Q Final Project: Demonstration of the Quantum Zeno Effect

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(Dated: June 11, 2026)

This is my wonderful abstract.

Introduction—

Theory—The quantum Zeno effect will be demonstrated here by considering the evolution between polarization states. The basis states in this case are $|H\rangle$ and $|V\rangle$. An arbitrary linear polarization state, then, can be written as

$$|\psi\rangle = \cos(\theta)|H\rangle + \sin\theta |V\rangle, \tag{1}$$

where θ is the angle of polarization.

We then aim to calculate the magnitude of the state that transmits through the polarizer. Allowing the photon to interact with the polarizer is effectively "measuring" the state of the photon, which causes its wavefunction to collapse into $|H\rangle$. Thus, in other words, we want to calculate the magnitude of state projected onto the polarizer axis. Without loss of generality, we will assume the polarizer is set to the $|H\rangle$ axis.[1] The projection operator [1] onto the $|H\rangle$ state is

$$P_H = |H\rangle\langle H|. \tag{2}$$

Thus, projecting the arbitrary linear polarization onto $|H\rangle$ gives

$$\begin{aligned} P_H|\psi\rangle &= |H\rangle\langle H|(\cos(\theta)|H\rangle + \sin\theta|V\rangle) \\ &= \cos(\theta)|H\rangle. \end{aligned} \tag{3}$$

The magnitude of this state is

$$\langle\psi|\psi\rangle = \cos^2(\theta), \tag{4}$$

which gives the probability that a given photon is transmitted. If there is a stream of photons entering a detector,

$$\langle\psi|\psi\rangle \propto \Gamma, \tag{5}$$

where Γ is the coincidence count rate. Thus, the relation between the input and output count rate is given by

$$\Gamma(\theta) = \Gamma_{\max}\cos^2(\theta). \tag{6}$$

For classical theories of light as an electromagnetic wave, this relation considering $\Gamma \leftrightarrow I$ is also known as Malus's law [2].

In this experiment, we wish to verify this relation is true for single photons as it traverses multiple polarizers, each collapsing the photon's wavefunction into its polarization direction. If there are n polarizers set so that the angle θ between them is consistent, this is equivalent to projecting successively onto different polarization states. Thus, the coincidence count rate for n polarizers with angle difference θ is

$$\Gamma(\theta, n) = \Gamma_{\max}\cos^{2n}(\theta). \tag{7}$$

In the case where the total angle difference is 90°, the relation between n and θ is

$$\theta = \frac{90^\circ}{n} = \frac{\pi}{2n}. \tag{8}$$

This leads to expected expression for the coincidence count rate for the experiment [3],

$$\Gamma(n) = \Gamma_{\max}\cos^{2n}\left(\frac{\pi}{2n}\right), \tag{9}$$

or equivalently,

$$\Gamma(\theta) = \Gamma_{\max}\cos^{\frac{\pi}{\theta}}(\theta). \tag{10}$$

It is critical to note that for increasing n, the rate of coincident photons also increases. Interestingly, for $n \to \infty$, $\Gamma \to \Gamma_{\max}$, meaning that all photons pass through despite the initial and final polarization angles being 90° apart. Thus, the quantum Zeno effect predicts that as more polarizers are added, more of the photons will transmit through, as the more frequent observations force wavefunction collapse in the polarizer bases.

Experimental Methods—

[1] If the polarizer set at a different angle α, this is equivalent to rotating the polarization state by $-\alpha$.

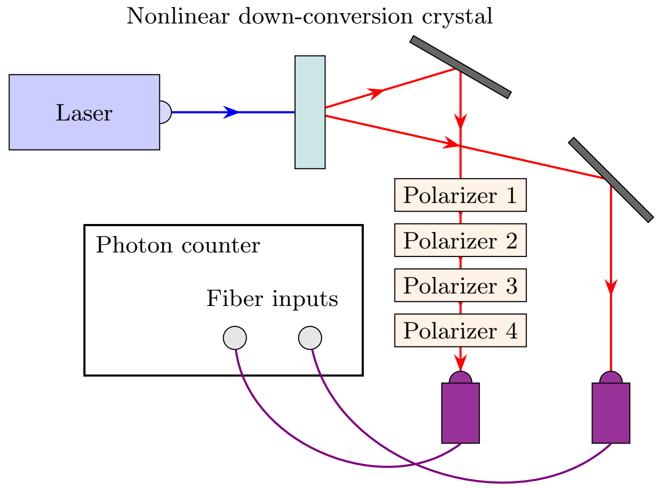

FIG. 1. Experimental setup. The laser diode outputs 405 nm horizontally-polarized light, which passes through a nonlinear down-conversion crystal (beta-barium borate, or BBO), which causes the incident photons to undergo spontaneous parametric down-conversion. This then generates pairs of horizontally-polarized 810 nm photons, which then are separated into two arms. The photon in Arm 0 passes through 4 successive polarizers, whereas the photon in Arm 1 passes through free space. At the end of each arm, the photons are coupled into optical fibers and led into the input of a photon counter that detects coincident photons.

Results—This experiment was performed for up to 4 polarizers, set at equal angles from 0° to 90°. The coincidence count rate was recorded for each configuration, with the baseline dark counts subtracted (Fig. 2). These data were fit to a modification Equation 9,

$$\Gamma(n) = A\cos^{2n}\left(\frac{\pi}{2n}\right) + C, \tag{11}$$

with A and C free fit parameters. The model showed a fair fit, with $R^2 = 0.9952$ and reduced $\chi^2 = 4.2447$. However, physical size constraints limited the number of data points possible.

Error analysis was performed assuming a Poisson distribution for the number of counts,

$$\sigma = \sqrt{N}, \tag{12}$$

and an error of 0.5° in the angles. The accidental count rate was also corrected for using the relation,

$$asdasd \tag{13}$$

The data fit to a linear function with free slope and intercept. The fitted slope is 224(39) counts/s/polarizer. The quantum Zeno effect-free case predicts a zero slope in this case, as increasing n does not increase the number of transmitted photons. However, we see the violation of the null hypothesis by $5.80\,\sigma$, a strong violation. Thus, we observe evidence of the quantum Zeno effect.

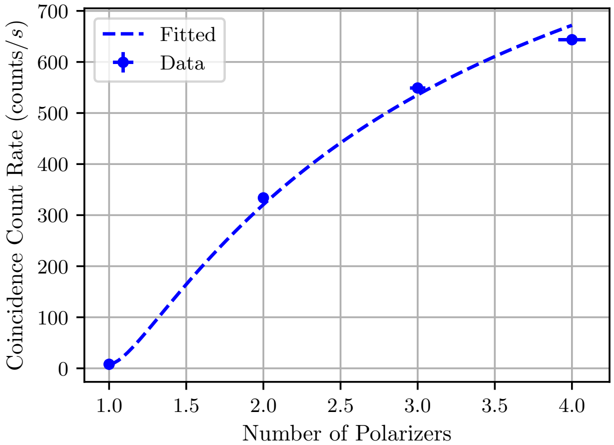

FIG. 2. Coincidence count rate as a function of number of polarizers.

Conclusion—

Summary—

[1] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, third edition ed. (Cambridge University Press, Cambridge, 2018).

[2] E. Hecht, *Optics*, 5th ed. (Pearson Education, Inc, Boston, 2017).

[3] P. Kwiat, H. Weinfurter, T. Herzog, A. Zeilinger, and M. Kasevich, Experimental Realization of Interaction-free Measurements, Annals of the New York Academy of Sciences **755**, 383 (1995).